		Target engagement	Dose	Efficacy	Safety
#32	**Hodgkins lymphoma** anti-PDCD1 antibodies (6)	**2014** Ex-vivo PD-1 binding on activated human T cells	**2015** Mouse syngeneic tumour PK-PD (MC38, Sa1/N) with ex-vivo binding EC50	**2018** 9p24.1 FISH on cHL diagnostic biopsies	**2016** Mouse allogeneic bone-marrow transplant GVHD… △ Silent on immune toxicity seen i…
	Hodgkins lymphoma anti-CD274 antibodies (3)	**2022** In-vitro PD-L1 binding and PD-1 blockade assays (engineered PD-L1/PD-1 cell lines, CD4+ T-cell…	**2022** Cynomolgus monkey repeat-dose toxicology (4- and 26-week GLP studies, NOAEL 200…	**2022** 9p24.1/PD-L1 genetics of… CS1001 limited despite 9p24.1 amplification	no model relied on
#33	**open-angle glaucoma** ROCK inhibitors (6)	**2018** Normotensive monkey eye, topical netarsudil 0.04%	**2014** Normotensive monkey eye, topical netarsudil 0.04%	**2014** New Zealand white rabbit, normotensive and acute… △ Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
	open-angle glaucoma PTGER2 agonists (3)	**2019** Cynomolgus monkeys with laser-induced ocular hypertension (plus normotensive rabbits and…	**2019** Cynomolgus monkeys with laser-induced ocular hypertension (plus normotensive rabbits and…	**2020** Cynomolgus monkeys with laser-induced ocular hypertension (plus normotensive rabbits and…	**2022** Cynomolgus monkey 13- and 39-week topical ocular GLP toxicity studies
#34	**age-related macular…** anti-VEGFA antibodies (5)	**2019** Mouse laser-induced CNV (brolucizumab)	**2020** Rabbit and dog intravitreal toxicology (abicipar) IOI 15% below stated margin	**2020** Mechanistic multi-compartment ocular PK model, rabbit-scaled to human (abicipar)	**2019** NZW rabbit intravitreal TK/tolerability (brolucizumab) △ Intraocular inflammation missed…
Allergy	**allergic disease** allergen (immunotherapy) (5)	**2019** Patient PBMC Fel d 1-specific CD4+ T cells (ex vivo), Cat-PAD	**2019** Allergen exposure chamber	**2018** Mouse airway allergy, Fel d 1 Mouse protection, field loss to placebo	**2019** Allergen exposure chamber
	allergic rhinitis PTGDR2 antagonists (4)	**2013** Human whole-blood PGD2 eosinophil shape-change assay	**2014** Asthmatic allergen challenge	**2014** Asthmatic allergen challenge PoM-backed regimen failed Phase 3 primary	**2017** Asthmatic allergen challenge
	allergic rhinitis allergen (immunotherapy) (8)	**2018** Conjunctival test, gp-ASIT+	**2019** Allergen exposure chamber	**2019** Conjunctival test, gp-ASIT+ Confirmatory field trial missed primary	**2019** Allergen exposure chamber
Autoimmune	**Sjogren syndrome** BTK inhibitors (3)	**2021** Rat BTK occupancy and collagen-induced arthritis (remibrutinib)	**2021** Semi-mechanistic PK/PD occupancy model (tirabrutinib) Not measured in patients	**2022** Rat BTK occupancy and collagen-induced arthritis (remibrutinib)	**2025** Rat and dog repeat-dose toxicology (remibrutinib)
	Sjogren syndrome PIK3CD inhibitors (3)	**2020** AdV5 cannulation focal sialadenitis mouse (seletalisib)	**2015** AdV5 cannulation focal sialadenitis mouse (seletalisib) No checkable prediction	**2019** Spontaneous NOD.ShiLtJ and NOD.B10 Sjögren's mice… No checkable prediction	**2023** Leniolisib rat and monkey repeat-dose toxicology
	psoriasis JAK1 inhibitors (3)	**2018** Human whole-blood cytokine-induced STAT phosphorylation, JAK1 selectivity	**2022** Rat, monkey repeat-dose tox	**2018** Rat adjuvant-induced arthritis (cross-indication; no psoriasis model was run)	**2022** Rat, monkey repeat-dose tox Dose-related platelet fall the animals missed
	psoriasis PDE4 inhibitors (3)	**2020** Human Th17 ex vivo skin model Not measured in patients	**2010** Mouse phorbol ear edema	**2019** In-vitro PDE4 enzyme and cellular assays, skin-delivery design △ Imiquimod mouse ran crisaborol…	**2011** Rabbit primary skin irritation and mouse local lymph node assay (2% ointment)
	psoriasis RORC inverse agonists (4)	**2021** Human whole-blood IL-17 secretion assay △ No IL-17 change in plaques	**2021** Human whole-blood IL-17 secretion assay	**2022** IL-23 minicircle mouse, oral BI 730357	**2015** Nonclinical toxicology species (not public) Model data unpublished
	psoriasis anti-IL17A antibodies (4)	**2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** Cynomolgus repeat-dose toxicology	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** People with inborn errors of IL-17 immunity
	psoriasis anti-IL23A antibodies (3)	**2023** Acute local intradermal human IL-23 mouse skin assay dosed with mirikizumab	**2018** Cynomolgus monkey repeat-dose toxicology (NOAEL)	**2019** Human psoriasis skin xenograft on AGR129 mice (anti-p19 clone 7G10)	**2018** Mouse intracellular-infection challenge with surrogate anti-p19 (BCG, Salmonella, Listeria)
	systemic lupus erythematosus BTK inhibitors (3)	**2021** Human PK/PD model of BTK inhibition fitted to fenebrutinib… △ Evobrutinib: anti-dsDNA…	**2021** Human PK/PD model of BTK inhibition fitted to fenebrutinib healthy-volunteer data	**2020** NZB/W F1 lupus mice (spontaneous or IFNα-accelerated… All three members failed SRI-4	**2017** Nonclinical GLP toxicology of member drugs (species not public) Model data unpublished
	ulcerative colitis S1PR1 agonists (3)	**2016** Rodent blood lymphocyte counts (ozanimod)	**2023** Rat and dog repeat-dose toxicology (etrasimod)	**2016** TNBS and CD4+CD45RBhi transfer colitis mice (ozanimod)	**2023** hERG current assay in vitro (etrasimod) △ Ozanimod still slows the heart
Endocrine	**endometriosis** GNRHR antagonists (3)	**2022** Rat HPG axis (LH in male rats; E2 in pro-oestrus female rats), linzagolix	**2022** Linzagolix PK-PD systems-pharmacology model	**2024** Linzagolix PK-PD systems-pharmacology model	**2022** Linzagolix PK-PD systems-pharmacology model △ Tox blind to BMD loss; liver held
	hypopituitarism GHR agonists (3)	**2016** Rhesus monkey PK/PD and repeat-dose toxicology (somatrogon)	**2015** Juvenile cynomolgus monkey (IGF-1 PK/PD) 5x monkey dose still below daily GH	**2017** Hypophysectomized rat (weight gain, IGF-1) Somavaratan inferior to daily GH	**2022** Rhesus monkey PK/PD and repeat-dose toxicology (somatrogon)
	pituitary dwarfism GHR agonists (3)	**2020** Somapacitan population PK/PD (IGF-I indirect-response) model from phase 1 data, simulati…	**2022** Rat and cynomolgus repeat-dose toxicology (somapacitan)	**2017** Hypophysectomized rat… Phase 3 failed non-inferiority to daily GH	**2022** Rat and cynomolgus repeat-dose toxicology (somapacitan)
	secondary hyperparathyroidism CASR positive modulators (4)	**2013** 5/6-nephrectomy uremic rat	**2017** Population PK/PD model of etelcalcetide (PTH-calcium homeostasis, CaSR cooperativity)	**2015** 5/6-nephrectomy uremic rat	**2023** Rat gastric emptying and marmoset emesis vs cinacalcet
	type 1 diabetes mellitus SLC5A2 inhibitors (4)	**2014** STZ-diabetic rat	**2008** STZ-diabetic rat No checkable prediction	**2018** STZ-diabetic rat	**2018** Diet-induced obese rat, insulin-replete, dapagliflozin, 38 days △ DKA 3-4x placebo, missed in rats
Gastroenterology	**exocrine pancreatic insufficiency** PNLIP replacement / gene therapy (3)	**2018** Pancreatic duct-ligated, embolized minipig (PEI model)	**2015** Pancreatic duct-ligated, embolized minipig (PEI model) an… No checkable prediction	**2016** Pancreatic duct-ligated dogs (Mayo Clinic, DiMagno), dosed wit… △ CFA far below PERT in CF…	**2017** Sponsor nonclinical toxicology (species not public) Model data unpublished
	GERD ATP4A blockers (3)	**2019** Stimulated gastric acid secretion in dogs	**2017** Stimulated gastric acid secretion in dogs No checkable prediction	**2019** Rat reflux oesophagitis vs esomeprazole	**2025** Rat repeat dosing: gastrin, emptying, microbiota Model data unpublished
	opioid-induced constipation OPRM1 antagonists (4)	**2011** Rat opioid-induced small-intestinal transit inhibition Not measured in patients	**2017** Rat GI-transit vs tail-flick analgesia dose window	**2019** Mouse morphine-inhibited intestinal transit and gastric emptying (axelopran)	**2017** Morphine-dependent rats (precipitated withdrawal)
	primary biliary cholangitis NR1H4 agonists (3)	**2024** Rodent FXR target-gene PD models	**2019** Rodent FXR target-gene PD models No checkable prediction	**2021** Mdr2-/- mice Patient result unclear	**2020** Mdr2-/- mice Ductular-reaction prediction never tested in PBC
Hematology	**hemophilia A** F8 replacement / gene therapy (8)	**2025** Cynomolgus macaques with pre-existing anti-AAV5 antibodies given valoctocogene roxaparvovec	**2017** F8-knockout mice + cynomolgus macaques	**2021** Hemophilia A dogs (Chapel Hill / Queen's colonies) △ Mouse-stable BAX 888 lost in man	**2023** Hemophilia A dogs (Chapel Hill / Queen's colonies) △ ALT rise in 81% of patients
	hemophilia B F9 replacement / gene therapy (7)	**2022** Rhesus macaques given AAVS3-hFIX vectors (FLT180a… △ FIX expression lost in 7 of 8	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua	**2024** Juvenile hemophilia B dogs, canine surrogate vector	**2024** Cynomolgus GLP tox, fidanacogene predecessor △ No such harm in patients
	PNH anti-C5 antibodies (4)	**2022** Hemolysis assay with C5 R885H variant	**2020** Cynomolgus PK/PD (crovalimab)	**2018** Complement hemolysis assay in human serum	**2020** Drug–target–drug complex assay, human C5
Metabolic	**Fabry disease** GLA replacement / gene therapy (4)	**2026** Gla-knockout Fabry mice given clinical-scale ST-920 (AAV2/6 liver-directed hGLA) in a 3-month…	**2023** Six-month IV toxicology, mice and cynomolgus monkeys (pegunigalsidase alfa)	**2024** NOD/SCID/Fabry (NSF) mice xenotransplanted with lentivirus (LV/AGA)-transduced human CD3…	**2026** Gla-knockout Fabry mice given clinical-scale ST-920 (AAV2/… △ Patients' liver enzymes rose;…
	glycogen storage disease II GAA replacement / gene therapy (3)	**2023** Human whole-blood / neutral-pH enzyme stability assay (cipaglucosidase alfa with miglustat)	**2023** Rat and cynomolgus monkey repeat-dose toxicology (exposure margin)	**2021** Gaa knockout mice (glycogen, muscle function)	**2023** Rat and cynomolgus monkey repeat-dose toxicology
	hyperlipidemia CETP inhibitors (3)	**2014** Human CETP/apoA-I transgenic mouse	**2014** CETP/apoA-I mouse; rat blood pressure △ Exposure far below prediction	**2011** Human CETP loss-of-function genotypes ☐ Outcome trial negative	**2014** Rat blood pressure + human adrenal H295R cells
	hyperlipidemia HCAR2 agonists (3)	**2015** Rat, guinea pig plasma NEFA	**2012** Rat plasma FFA suppression (Merck screen) △ NEFA suppression not held	**2017** Rat, guinea pig plasma NEFA GSK256073 phase 2a stopped for futility	**2017** Guinea pig ear-temperature flush model (GSK)